

William Zhu

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Education

Carleton University, Ottawa, ON

May 2026

- GPA: 3.9/4.0, Honours Bachelor of Computer Science, Minor in Psychology

Experience

Cloud Engineer Intern, Kinaxis – Ottawa, ON

Jan 2025 – Aug 2025

- Integrated Slim CI/CD pipelines for DBT model deployment in BigQuery, leveraging model-lineage tracking and unit testing, reducing build costs by 67%, accelerating runtimes by up to 98%, and saving an estimated \$39,722.95 annually in storage.
- Designed and automated ETL pipelines with DLT and Python, ingesting data from backend APIs (ServiceNow, Datadog, internal servers) and orchestrating workflows via GitHub Actions to deliver FinOps analyses, dashboards, and ML-driven insights.
- Developed and productionized data models using DBT, SQL, and Python, implementing complex data transformations and formatting to produce metrics and gold-standard datasets for trend analysis, volume tracking, and AI applications
- Created and maintained several production-level Power BI semantic models and reports using financial, HR, and performance data from dataflows to compute key metrics and deliver data insights

DevOps Intern, OpenText – Ottawa, ON

May 2024 – Aug 2024

- Developed an IaC monitoring stack for cloud expenses with automated deployments using Pulumi, AWS, Boto3, Docker, and Apache Superset, tracking and rapidly identifying financial anomalies, reducing expenditures by 26% and prevented an actual leakage of thousands of dollars
- Implemented a CI/CD pipeline for the monitoring stack using a three-tiered approach for testing and releasing deployments to efficiently scale resources and minimize idle times
- Scripted a CI job using Bash and YAML that dynamically generated and deployed CI/CD pipelines executing Trivy scans to identify security vulnerabilities in third-party Docker images

Cloud Architecture Intern, Communications Research Centre Canada – Ottawa, ON

Sep 2023 – Dec 2023

- Automated deployment with IaC of a data processing sequence using Python, AWS CDK, AWS CLI, and various AWS services, improving deployment speed by 83%
- Developed a Python script using AWS CDK to process, store, and visualize CSV data from an Amazon S3 bucket into a filterable graphical format using Amazon QuickSight templates, analyses, datasets, and dashboards
- Created modular Amazon QuickSight templates using AWS CLI and Amazon QuickSight to minimize configuration and enable rapid redeployment across different datasets

Junior Programmer, Canada Revenue Agency – Ottawa, ON

May 2023 – Sep 2023

- Implemented audit processing feature to display new audit information and condition specific warning messages
- Detected program vulnerabilities and optimized functionality through black-box, white-box, and regression testing to improve overall program robustness

Projects

D&D Spell Index and Calculator 

- Built a cloud-based D&D spell database using AWS Lambda, AWS RDS, Python, and SQL, enabling menu-based queries that return spell descriptions and computed mechanics in a neat human-readable format.

Predictive Depression Classification Ensemble 

- Developed an end-to-end data processing pipeline and machine learning ensemble integrating Random Forest and XGBoost, performing input cleaning, feature partitioning, and weighted-voting classification to predict depression in unseen patients.

Skills

Cloud Services: (GCP) BigQuery, Bucket, Cost Explorer, (AWS) Lambda, RDS, Cost Explorer, S3, Quicksight, CloudTrail, CloudWatch, EC2

Technologies: DBT, DLT, AWS CDK, AWS CLI, Pulumi, Boto3, Express.js, Node.js, Pandas, Scikit-Learn, Numpy

Tools: Power BI, Github Actions, Gitlab CI/CD, Datadog, Docker, PostgreSQL, Apache Superset, ServiceNow, Grafana, MongoDB, Git

Languages: Python, SQL, YAML, Bash, Java, C++, C, JavaScript, HTML/CSS